

FMF & Autoinflammatory Syndromes — The Gallery (VEXAS, PAPA, DADA2, Blau & friends)





1. VEXAS — UBA1

WHAT YOU SEE

Central portrait of older man, VEXAS, UBA1 gene

WHAT IT ENCODES

VEXAS syndrome is caused by an ACQUIRED SOMATIC mutation in UBA1 (the E1 ubiquitin-activating enzyme) in myeloid progenitor cells. Because UBA1 is on the X chromosome, it almost exclusively affects older MEN (>50). The acronym: Vacuoles (in myeloid/erythroid precursors on marrow biopsy), E1 enzyme (UBA1), X-linked, Autoinflammatory, Somatic. Described in 2020, it bridges rheumatology and hematology.

BOARD TRAP

VEXAS is an ACQUIRED somatic mutation of adulthood — NOT an inherited childhood autoinflammatory syndrome. The trap is treating an older man's refractory inflammation as a benign autoimmune condition and missing the bone marrow / UBA1 workup.

2. VEXAS features

WHAT YOU SEE

Scattered vasculitis/nodules, fever, cytopenias, MDS link

WHAT IT ENCODES

VEXAS presents with treatment-refractory systemic inflammation — recurrent fevers, chondritis (ear/nose, relapsing-polychondritis-like), skin vasculitis/neutrophilic dermatosis, periorbital/orbital inflammation, lung infiltrates — PLUS hematologic disease: macrocytic anemia, progressive cytopenias, and a strong association with myelodysplastic syndrome (MDS) and VTE. It is often steroid-dependent and tends to be poorly responsive to standard immunosuppression; azacitidine or allogeneic transplant may be needed.

BOARD TRAP

Suspect VEXAS in an older man with refractory inflammation AND unexplained macrocytic anemia / cytopenias / MDS — the wrong path is to keep escalating empiric immunosuppression for presumed seronegative rheumatic disease without marrow biopsy and UBA1 sequencing.

3. PAPA syndrome

WHAT YOU SEE

Teen with leg ulcer, PAPA, PSTPIP1, 'sterile pyogenic'

WHAT IT ENCODES

PAPA = Pyogenic (sterile) Arthritis, Pyoderma gangrenosum, and Acne. It is autosomal DOMINANT due to PSTPIP1 mutations that dysregulate pyrin/inflammasome signaling, presenting in childhood with recurrent destructive sterile arthritis that later gives way to severe cystic acne and pyoderma gangrenosum. IL-1 and TNF inhibitors are used.

BOARD TRAP

PAPA arthritis is STERILE despite looking pyogenic — synovial fluid is purulent-appearing but cultures are NEGATIVE. Treating it as septic arthritis with antibiotics alone (and missing the autoinflammatory diagnosis) is the trap.

4. DADA2

WHAT YOU SEE

Child with livedo/purple skin, DADA2, ADA2 / 'early strokes'

WHAT IT ENCODES

DADA2 = Deficiency of ADA2, autosomal recessive loss-of-function in ADA2/CECR1. It causes a PAN-like medium-vessel vasculitis with early-onset (childhood) recurrent ischemic/hemorrhagic LACUNAR STROKES, livedo racemosa, fevers, and variable immunodeficiency/cytopenias (hypogammaglobulinemia, pure red cell aplasia).

BOARD TRAP

DADA2 causes EARLY childhood strokes + livedo — the correct therapy is ANTI-TNF (which prevents strokes), NOT antiplatelets/anticoagulation alone, and anticoagulation can be dangerous given hemorrhagic risk. Treating it as ordinary pediatric stroke or isolated PAN is the trap.

5. Blau syndrome

WHAT YOU SEE

Child with swollen joints, BLAU, NOD2, AD

WHAT IT ENCODES

Blau syndrome is autosomal DOMINANT NOD2 (CARD15) gain-of-function, presenting in early childhood (<4 yrs) with the triad of granulomatous polyarthritis (boggy synovitis/tenosynovitis), granulomatous uveitis (which can threaten vision), and granulomatous dermatitis. Histology shows noncaseating granulomas.

BOARD TRAP

Blau is early-onset GRANULOMATOUS arthritis/uveitis/rash from NOD2 — it looks like sarcoidosis but is a monogenic childhood disease (early-onset sarcoidosis is the same entity). The trap is diagnosing childhood sarcoid without recognizing the NOD2 genetics. Note NOD2 loss-of-function is linked to Crohn's, while Blau is gain-of-function.

6. Majeed / SAVI / interferonopathies

WHAT YOU SEE

INTERFERONOPATHIES panel, SAVI rash, type I IFN

WHAT IT ENCODES

Type I interferonopathies (e.g., SAVI from STING gain-of-function; CANDLE; Aicardi-Goutières) are driven by excess type I interferon (IFN-α/β) rather than IL-1. SAVI causes early-onset systemic inflammation, severe acral/facial vasculopathy with skin ulceration and digital gangrene, and interstitial lung disease.

BOARD TRAP

Interferonopathies are IFN-driven — the rational therapy is a JAK inhibitor (baricitinib/ruxolitinib) to block IFN signaling, NOT IL-1 blockade. Reaching for anakinra in an IFN-signature disease is the wrong-mechanism trap.

7. Rosai-Dorfman / histiocytosis

WHAT YOU SEE

ROSAI panel, massive nodes/skin lesions

WHAT IT ENCODES

Histiocytic disorders sit in the systemic-inflammation differential. Rosai-Dorfman disease (sinus histiocytosis with massive lymphadenopathy) features painless massive cervical lymphadenopathy and infiltrative skin/extranodal lesions; biopsy shows emperipolesis (intact lymphocytes within histiocytes) and S100-positive, CD1a-negative histiocytes.

BOARD TRAP

Massive painless lymphadenopathy with EMPERIPOLESIS on biopsy points to Rosai-Dorfman, not FMF or another periodic fever. The trap is anchoring on 'recurrent fever + nodes' and missing the histiocytic diagnosis that biopsy reveals.



8. CRMO / chronic osteomyelitis

WHAT YOU SEE

CRMO panel, bone pain figure

WHAT IT ENCODES

Chronic recurrent multifocal osteomyelitis (CRMO) is a sterile autoinflammatory bone disease of children/adolescents with multifocal, often metaphyseal, lytic/sclerotic lesions and bone pain; it falls within the SAPHO spectrum (Synovitis, Acne, Pustulosis, Hyperostosis, Osteitis). NSAIDs are first-line; bisphosphonates/TNF inhibitors for refractory disease.

BOARD TRAP

CRMO bone lesions are STERILE — cultures are negative and ANTIBIOTICS do not help. The trap is repeatedly treating it as bacterial osteomyelitis; it is autoinflammatory, and biopsy/cultures distinguish it from infection.

9. Schnitzler syndrome

WHAT YOU SEE

SCHNITZLER panel, urticaria + monoclonal IgM

WHAT IT ENCODES

Schnitzler syndrome is an adult-onset acquired autoinflammatory disorder defined by chronic urticarial rash PLUS a monoclonal IgM gammopathy (rarely IgG), with recurrent fever, bone pain, arthralgia, lymphadenopathy, and elevated inflammatory markers. It is IL-1β driven and responds dramatically to anakinra; there is a long-term risk of progression to Waldenström macroglobulinemia/lymphoproliferative disease.

BOARD TRAP

Schnitzler = chronic urticaria + monoclonal IgM in an ADULT — order SPEP/immunofixation to find the IgM paraprotein. The trap is treating the rash as ordinary chronic idiopathic urticaria with antihistamines and missing the monoclonal gammopathy; it responds to IL-1 blockade, not antihistamines.

10. MEN / inheritance tags

WHAT YOU SEE

X-linked / AD / AR labels under each portrait

WHAT IT ENCODES

Inheritance pattern is a high-yield sorting tool across the gallery: VEXAS is X-linked AND acquired/somatic; PAPA and Blau are autosomal DOMINANT; DADA2 is autosomal RECESSIVE; FMF and HIDS are recessive while TRAPS is dominant.

BOARD TRAP

Boards test inheritance directly — don't mix up the dominant syndromes (PAPA, Blau, TRAPS) with the recessive ones (DADA2, FMF, HIDS). Calling DADA2 dominant or VEXAS inherited-germline is the classic trap.

11. Sterile inflammation theme

WHAT YOU SEE

'Sterile pyogenic' framing across the gallery

WHAT IT ENCODES

The unifying theme: autoinflammatory disease is driven by the INNATE immune system (inflammasome/cytokine dysregulation, especially IL-1 and IFN) WITHOUT high-titer autoantibodies or antigen-specific T cells. Inflammation is sterile, recurrent/stereotyped, and culture-negative.

BOARD TRAP

Autoinflammatory is NOT the same as autoimmune — there are no characteristic autoantibodies (no ANA/RF/anti-dsDNA driving it) and complement/serologies are typically unremarkable. Ordering an autoantibody panel and expecting it to clinch the diagnosis is the conceptual trap.

12. Treatment cheat sheet

WHAT YOU SEE

Bench placard listing IL-1 / anti-TNF / JAK options

WHAT IT ENCODES

Targeted therapy follows the cytokine driver: IL-1 blockers (anakinra/canakinumab) for cryopyrinopathies, Schnitzler, and refractory FMF; anti-TNF for DADA2 (stroke prevention) and Blau; JAK inhibitors for type I interferonopathies (SAVI/CANDLE); colchicine for FMF. VEXAS is often refractory and may need azacitidine or transplant.

BOARD TRAP

Match the biologic to the cytokine driver — a wrong target equals treatment failure. The trap is reflexively choosing IL-1 blockade for everything: it fails in interferonopathies (need JAK) and is not the answer for DADA2 (anti-TNF) or VEXAS.